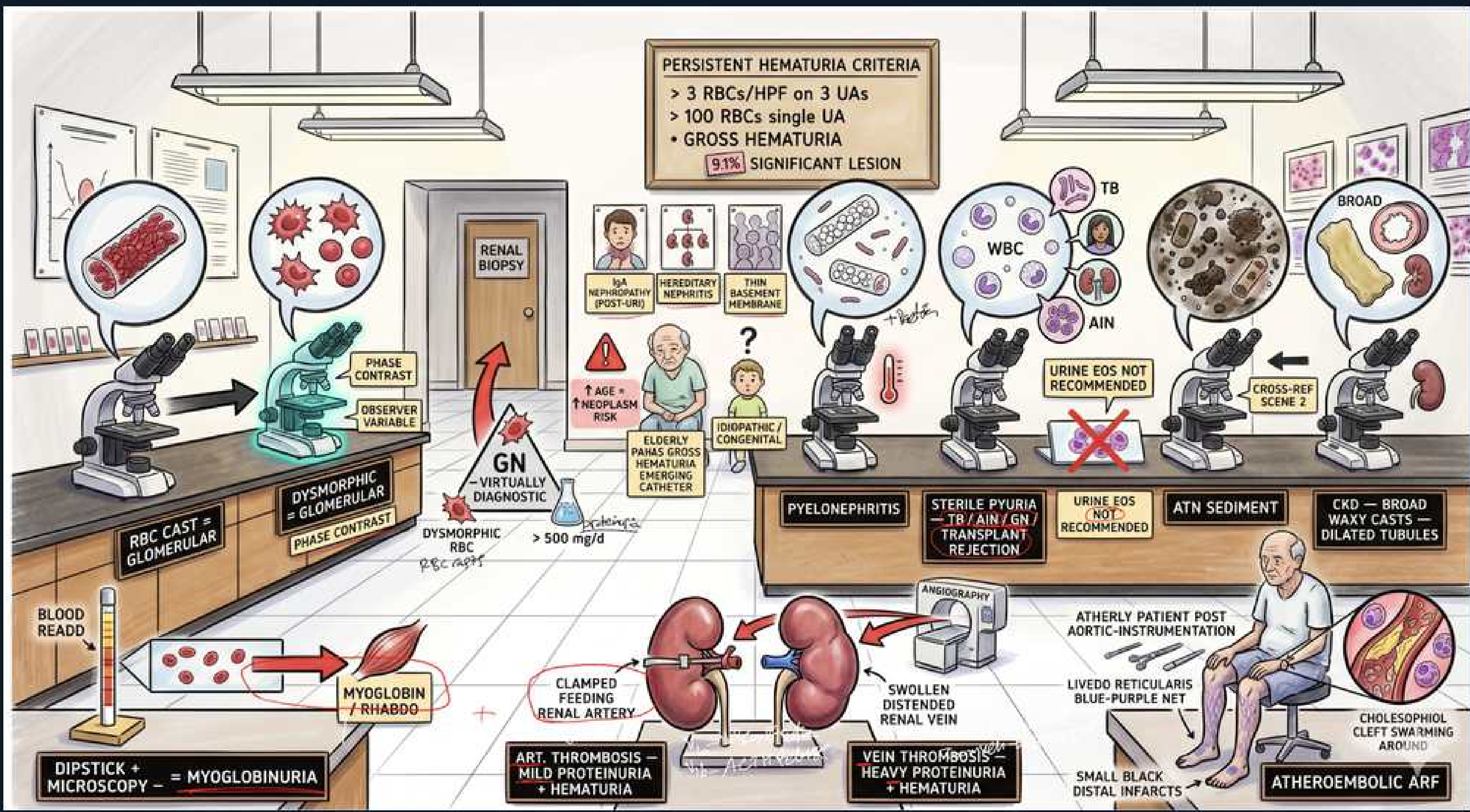
Hematuria & Atheroembolic ARF — Localizing the Bleed



1. Persistent hematuria criteria

WHAT YOU SEE

Sign: >3 RBC/HPF on 3 UAs, or >100 RBC single UA, or gross hematuria.

WHAT IT ENCODES

Microscopic hematuria is defined as >=3 RBCs/HPF on a properly collected UA, confirmed on repeat. Gross or high-count single samples and any visible blood warrant evaluation. ~5% of persistent hematuria harbors a significant urologic lesion (malignancy).

BOARD TRAP

A single positive dipstick is NOT hematuria — dipstick detects heme and is falsely positive with myoglobin/hemoglobin; you must see RBCs on micro.

2. RBC cast = glomerular

WHAT YOU SEE

Microscope with red tube-shaped cast.

WHAT IT ENCODES

RBC casts form in the tubule and are pathognomonic for glomerular bleeding (glomerulonephritis). Localizes the source ABOVE the bladder/ureter. Drives you to nephrology workup, not cystoscopy.

BOARD TRAP

RBC casts point to GN, not stones or tumor — don't order cystoscopy first when casts are present.

3. Dysmorphic RBC / phase contrast

WHAT YOU SEE

Phase-contrast microscope, distorted RBCs.

WHAT IT ENCODES

Dysmorphic (acanthocyte/Mickey-mouse) RBCs indicate glomerular origin; >5% acanthocytes is highly specific. Phase-contrast microscopy improves detection. Observer-variable and requires fresh sample.

BOARD TRAP

Isomorphic (uniform) RBCs suggest a lower urologic source (stone/tumor) — shape, not just count, localizes the bleed.

4. Dysmorphic + >500 mg proteinuria = GN virtually diagnostic

WHAT YOU SEE

Red arrow 'GN virtually diagnostic, >500 mg/d'.

WHAT IT ENCODES

Dysmorphic RBCs PLUS significant proteinuria (>500 mg/day) make glomerulonephritis essentially certain and justify renal biopsy. Combination beats either finding alone.

BOARD TRAP

Hematuria with heavy proteinuria is glomerular until proven otherwise — isolated hematuria without proteinuria more often urologic.

5. Renal biopsy

WHAT YOU SEE

Door labeled 'RENAL BIOPSY'.

WHAT IT ENCODES

Biopsy confirms glomerular pattern when serologies/imaging are inconclusive and there is dysmorphic hematuria with proteinuria or rising creatinine. Defines the specific glomerulonephritis to guide immunosuppression.

BOARD TRAP

Don't biopsy isolated low-grade microscopic hematuria with normal renal function and no proteinuria — observe instead.

6. Myoglobin / rhabdomyolysis

WHAT YOU SEE

Vial labeled 'MYOGLOBIN / RHABDO', dipstick + micro = myoglobinuria.

WHAT IT ENCODES

Dipstick heme-positive but NO RBCs on micro = myoglobinuria (rhabdo) or hemoglobinuria. CK markedly elevated, risk of pigment-induced AKI; treat with aggressive IV fluids.

BOARD TRAP

Tea-colored urine + positive dipstick + absent RBCs = pigment, not hematuria — checking only the dipstick misses rhabdo.

7. Age >40 / urothelial CA risk

WHAT YOU SEE

Elderly patient, indwelling catheter cue.

WHAT IT ENCODES

Age >40, smoking, gross hematuria, and analgesic/chronic catheter use raise urothelial malignancy risk; these patients need cystoscopy + upper-tract imaging (CT urogram). ~5% significant lesion rate justifies full urologic workup.

BOARD TRAP

Gross painless hematuria in an older smoker is bladder cancer until excluded — do cystoscopy even if a UTI coexists.

8. Pyelonephritis

WHAT YOU SEE

Sign 'PYELONEPHRITIS', sterile pyuria note.

WHAT IT ENCODES

WBCs/WBC casts with bacteriuria and flank pain indicate pyelonephritis. Hematuria can accompany infection. Sterile pyuria (WBCs, negative culture) suggests TB, chlamydia, or interstitial disease.

BOARD TRAP

Sterile pyuria is NOT contamination — think renal TB, partially treated UTI, or analgesic nephropathy.

9. Urine EOS not recommended

WHAT YOU SEE
Sign 'URINE EOS NOT RECOMMENDED'.

WHAT IT ENCODES
Urine eosinophils are no longer recommended for acute interstitial nephritis (AIN) — poor sensitivity and specificity. AIN is suggested by drug exposure, rash, eosinophilia, and pyuria; biopsy confirms.

BOARD TRAP
Absent urine eosinophils does NOT rule out AIN, and their presence does not confirm it — the test was abandoned for low accuracy.

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10. ATN sediment

WHAT YOU SEE
Sign 'ATN SEDIMENT', muddy-brown casts.

WHAT IT ENCODES
Acute tubular necrosis shows muddy-brown granular casts and renal tubular epithelial cells. FeNa typically >2% (vs <1% in prerenal). Distinguishes intrinsic ATN from prerenal azotemia.

BOARD TRAP
Muddy-brown granular casts = ATN, not glomerular — don't confuse with RBC casts; FeNa >2% supports ATN over prerenal.

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11. CKD — broad waxy casts / dilated tubules

WHAT YOU SEE
Sign 'CKD — broad waxy casts, dilated tubules'.

WHAT IT ENCODES
Broad waxy casts form in dilated, atrophic tubules of advanced CKD and signify chronicity. Their presence implies long-standing disease and poor reversibility.

BOARD TRAP
Broad waxy casts indicate chronic, not acute, injury — seeing them shifts you away from a fully reversible AKI diagnosis.

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12. Atheroembolic ARF — livedo, blue toes

WHAT YOU SEE
Seated patient post-instrumentation, livedo reticularis, blue toes.

WHAT IT ENCODES
Cholesterol (atheroembolic) ARF follows aortic instrumentation/anticoagulation: livedo reticularis, blue toes with intact pulses, Hollenhorst plaques, eosinophilia, and low complement. Stepwise rising creatinine over weeks.

BOARD TRAP
Contrast nephropathy peaks at 3–5 days then recovers; atheroembolic disease climbs stepwise with skin findings + eosinophilia — don't blame the dye.

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13. Renal artery thrombosis

WHAT YOU SEE
Clamped feeding renal artery, 'mild proteinuria + hematuria'.

WHAT IT ENCODES
Acute renal artery thrombosis/embolism causes flank pain, LDH spike, and infarction with mild proteinuria + hematuria. Sources include AF, endocarditis, trauma. CT angiography confirms.

BOARD TRAP
Sudden flank pain with a sky-high LDH and AF points to renal infarction, not a passing stone — image the vasculature.

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14. Renal vein thrombosis

WHAT YOU SEE
Swollen distended kidney, 'heavy proteinuria + hematuria'.

WHAT IT ENCODES
Renal vein thrombosis classically complicates nephrotic syndrome (esp. membranous) — heavy proteinuria, flank pain, swollen kidney. Hypercoagulable from urinary antithrombin loss.

BOARD TRAP
New flank pain + hematuria in a nephrotic (membranous) patient is renal vein thrombosis — anticoagulate, don't just treat 'a UTI'.

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15. Cholesterol clefts in tissue

WHAT YOU SEE
Biopsy showing cholesterol clefts.

WHAT IT ENCODES
Biopsy of atheroembolic disease shows biconvex cholesterol clefts in arterioles. Confirms the embolic source when skin/eosinophil clues are present. Generally no specific therapy beyond risk-factor control.

BOARD TRAP
Cholesterol clefts confirm atheroemboli — steroids/anticoagulation are NOT proven treatment and anticoagulation may worsen embolization.
